

The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

A more accurate surrogate can be a worse RL training environment — it exposes a rougher action→temperature surface that the policy exploits into failure on the real building. A role-separating hybrid keeps the surface smooth and the controller robust.

Surrogate training environment

Coarse black-box v3
(1 h step — less accurate)

Accurate twin
(calibrated v3.5 / fine-res v3)

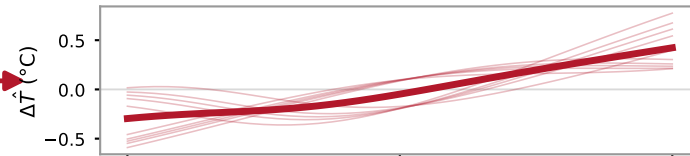
Hybrid
(v3 rollout + frozen
v3.5 censor)

Action → next-temperature response surface (measured)

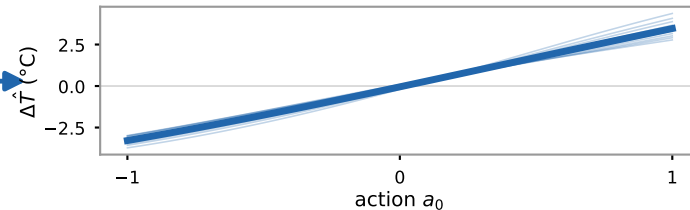
smooth · monotone



rough · non-monotone

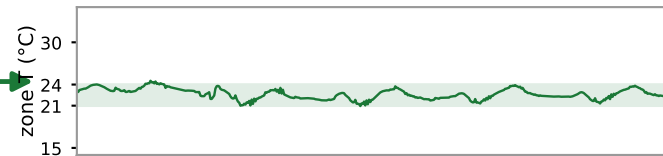


*smooth (v3 dynamics)
+ plausibility censor*

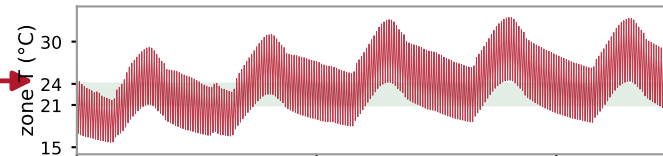


Zone temperature on the live building (measured)

✓ **USABLE**



✗ **COLLAPSE**



✓ **ROBUST**

